

Vangmayee Avadhanula

Data Analyst

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SUMMARY

Data Analyst with **3+ years** of experience turning complex data into actionable insights that improve business performance and operational efficiency. Strong in **SQL/PL/SQL, Python, Excel, Tableau, and Power BI**, with hands-on expertise in data cleaning, wrangling, mining, and statistical analysis using **Pandas, NumPy, SciPy, and scikit-learn**. Experienced delivering KPI reporting and stakeholder-ready dashboards across **AWS, Azure, and GCP**, and working with relational databases including **SQL Server, Oracle, PostgreSQL, and MySQL**. Known for optimizing data workflows and delivering data-driven recommendations that strengthen business outcomes.

SKILLS

Methodologies	SDLC, Agile, Waterfall
Programming Languages	Python, R, SAS, SQL, PL/SQL
IDEs	Visual Studio Code, PyCharm, Jupyter Notebook
Packages	NumPy, Pandas, SciPy, Matplotlib, Seaborn, Scikit-learn, TensorFlow, ggplot2, PySpark
ETL Tools	SSIS, SSRS, SSAS, Informatica, Azure Data Factory (ADF), AWS Glue
Visualization Tools	Tableau, Power BI, Excel (VLOOKUP, HLOOKUP, Pivot Tables)
Cloud Technologies	AWS (S3, Redshift, Glue, Athena, EMR, Lambda, Kinesis, QuickSight), Azure (Synapse, Data Factory, Databricks, Data Lake Storage, Cosmos DB, Stream Analytics), GCP (BigQuery, Dataflow, Pub/Sub)
Data Platforms & Warehousing	Snowflake, Databricks, Delta Lake, AWS Redshift, Azure Synapse, Google BigQuery, Microsoft Fabric
Databases	MySQL, PostgreSQL, SQL Server, Oracle, MongoDB
Version Control Tools	Git, GitHub, GitLab, SVN

EXPERIENCE

National Cooperative Bank	Jan 2025 - Present
Data Analyst	Arlington, VA

- Delivered KPI and fraud reporting over **5M+ transactions/month** for Risk & Compliance, enabling faster decisions and contributing to a **9% reduction in fraud losses** through weekly trend insights and action tracking.
- Standardized and reconciled KPI datasets (loan disbursements, CASA growth, NPA movement, transaction volume) by applying validation rules and exception checks across **30+ source tables**, improving report accuracy **15%** and reducing reconciliation issues **30%**.
- Built a rule-based fraud scoring model in **Python (Pandas, NumPy)** using statistical thresholds across **10M+ rows**, improving detection accuracy **12%** and identifying customer-level anomaly patterns.
- Developed and optimized **Power BI** dashboards (fraud rate, chargebacks, suspicious merchants, anomaly trends) using **DAX** and tuned SQL, reducing dashboard load time by **30%** for business users.
- Migrated legacy reporting datasets to **Snowflake (Azure-integrated)** and redesigned schemas/SQL to improve query performance **35%** and reduce compute costs **20%**; implemented incremental processing (partitioning/clustering) to cut batch time from **3h to 1.5h**.
- Operationalized daily KPI reporting (weekly to daily) using **Microsoft Fabric orchestration** and scheduled refreshes; partnered with Risk/Compliance on KPI definitions, UAT signoffs, and audit-ready delivery, reducing manual reporting effort **60%**.

Unisys Global Services	Jan 2022 – Jan 2024
Data Analyst	India

- Delivered claims and fraud BI reporting across **7M+ claim records** and **30+ operational KPIs**, enabling faster decisions and improving accountability through weekly performance readouts.
- Cleaned, normalized, and integrated multi-source claims and customer datasets using SQL transformations and prep layers, reducing data defects by **20%** and improving report reliability for leadership consumption.
- Built Python-based validation and anomaly/outlier checks across **10M+ rows**, improving fraud/risk analytics reliability and reducing false escalations by **15%**.
- Built **Tableau** dashboards for claims lifecycle, backlog trends, and risk segmentation using advanced calculations (including cumulative and trend metrics), reducing manual Excel reporting by **70%** and improving self-serve insights for **15+ stakeholders**.
- Queried and modeled analytics datasets on **AWS Redshift** and **GCP BigQuery**, optimizing SQL and dataset design to reduce refresh/processing time by **30%** and support scalable reporting for **15M+ records**.
- Deployed automated report distribution and alerting workflows (integrated with **AWS/GCP** reporting and Tableau extracts), collaborating with Ops, Fraud, and QA teams to define requirements, validate outputs, and roll out changes with documented metrics and governance.

CERTIFICATIONS

Microsoft Certified: Power BI Data Analyst Associate (PL-300)

PROJECTS

Author Bibliography & Book Sales Analysis.

- Analyzed a 4-year author bibliography dataset (10,000+ books) to identify trends in monthly sales, revealing consistent peaks with sales increasing by 35% from June–August (academic year) and 40% in December (holiday gifting).
- Applied NLP techniques to preprocess book titles and extract keywords correlated with higher sales, finding that titles containing words like “guide,” “adventure,” or “story” had 20–25% higher purchase rates.
- Conducted data cleaning, aggregation, and exploratory analysis using Python and Excel, generating actionable insights on sales patterns and title optimization.
- Visualized results through charts and dashboards, helping recommend seasonal marketing strategies and optimal book release timing.

Analyzing Traffic Stops and Incidents: Socioeconomic Associations

- Performed high-volume data mining on 1M+ traffic stop records to identify systemic disparities, utilizing Pearson Correlation and Chi-square tests to quantify links between race, gender, and search rates.
- Engineered feature sets incorporating socioeconomic indicators like median income and employment status to predict high-incident zones.
- Developed dynamic Tableau dashboards with spatio-temporal filters and Heat Maps, enabling stakeholders to visualize geographic “hot-spots” and trend shifts over 5-year intervals.
- Built a Random Forest classification model (88% accuracy) using SQL and Scikit-learn to identify high-risk regions, providing actionable evidence for bias-reduction policy making.

Cloud-Scale Analysis of Zomato Bangalore Dataset

- Architected a cloud-based analytics framework using AWS S3 and Athena to enable serverless SQL analysis over 50,000+ restaurant records, reducing ad-hoc query turnaround by 60% and lowering compute costs by 30%.
- Leveraged PySpark on Databricks to perform distributed sentiment analysis on 300,000+ customer reviews, achieving a 5 times improvement in processing throughput and producing sentiment scores with 0.82 F1 against labeled samples.
- Designed an end-to-end ETL pipeline to clean unstructured metadata, handling 20+ edge cases, increasing data completeness from 78% to 95%, and standardizing pricing and location features across 30+ Bangalore districts for market saturation analysis.
- Generated multivariate visualizations using Seaborn to model pricing elasticity across 10+ features, identifying a 0.65 correlation between specialty cuisines and premium customer satisfaction scores and surfacing three high-opportunity expansion zones.

EDUCATION

George Mason University
Master of Science in Data Analytic Engineering

Fairfax, VA
Dec 2025

Osmania University
Bachelor of Engineering in Electronics and Communication Engineering

Hyderabad, India
July 2023